Bahria University, Islamabad

Department of Software Engineering

Repository Link:
https://github.com/YOAmirAli/SQA-Search-Library

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Enrollment: 01-131232-011 & 093

Software Quality Assurance (Spring-2026)

Instructor: Dr. Tamim Ahmed Khan

Assignment: 01

Date: 1/3/2026

Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1					
2					
3					
4					
5					

Comments:

Assignment 1: Software Quality Engineering (SQE-6)

1. Project Overview

The Software Under Test (SUT) is a .NET 8.0 library named SearchAlgorithmsLibrary. This project implements and verifies core search algorithms, including Linear Search, Jump Search, and a specifically verified Binary Search. The project objective was to apply xUnit testing frameworks and achieve high-quality standards through Modified Condition/Decision Coverage (MC/DC) and automated CI/CD pipelines via GitHub Actions.

2. Integration Strategy

We followed a Bottom-Up Integration strategy to ensure robust verification:

- Unit Level: Individual methods were tested locally using xUnit to catch logic errors early.
- Integration Level: Code was pushed to GitHub to trigger an automated build and test sequence, ensuring the library correctly interacts with the test project environment.

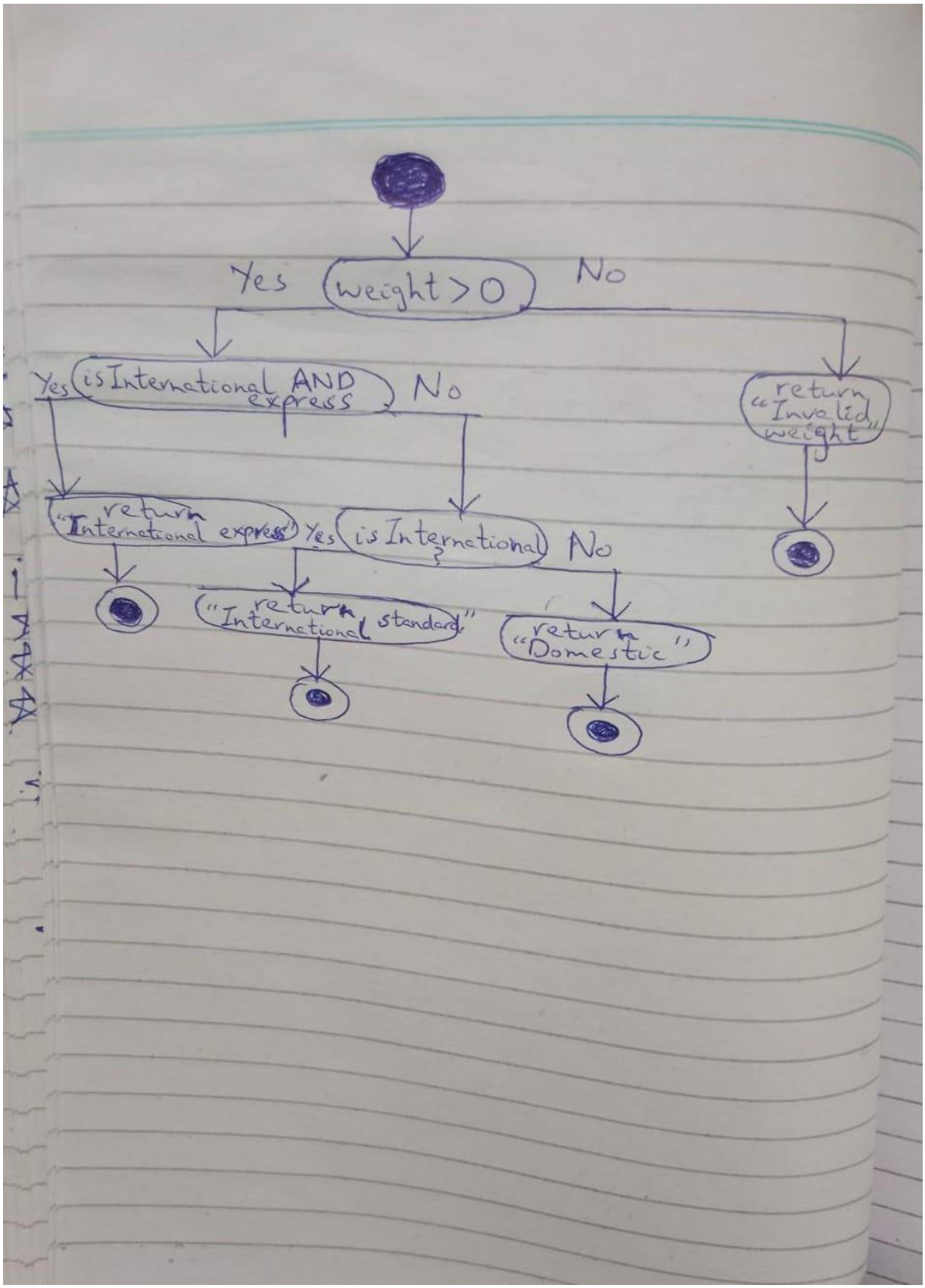
3. MC/DC Testing Analysis

To fulfill the requirement for high-integrity testing, we applied Modified Condition/Decision Coverage (MC/DC) to the shipping logic.

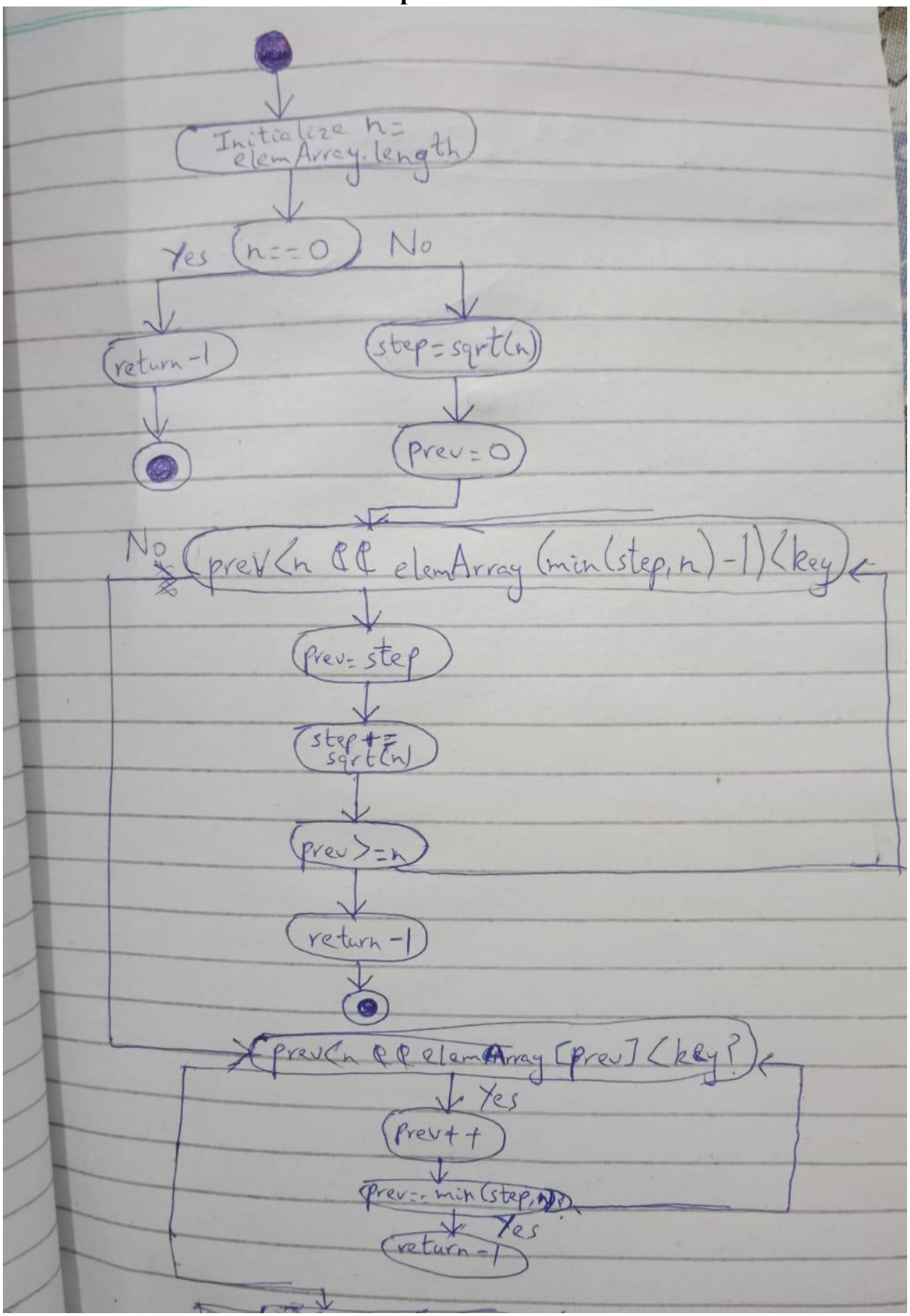
Test Case	**weight > 0**	**isInternational**	**express**	**Expected Output**	**Actual Result**
TC1	True	True	True	"International Express"	Pass
TC2	True	True	False	"International Standard"	Pass
TC3	True	False	(any)	"Domestic"	Pass
TC4	False	(any)	(any)	"Invalid Weight"	Pass

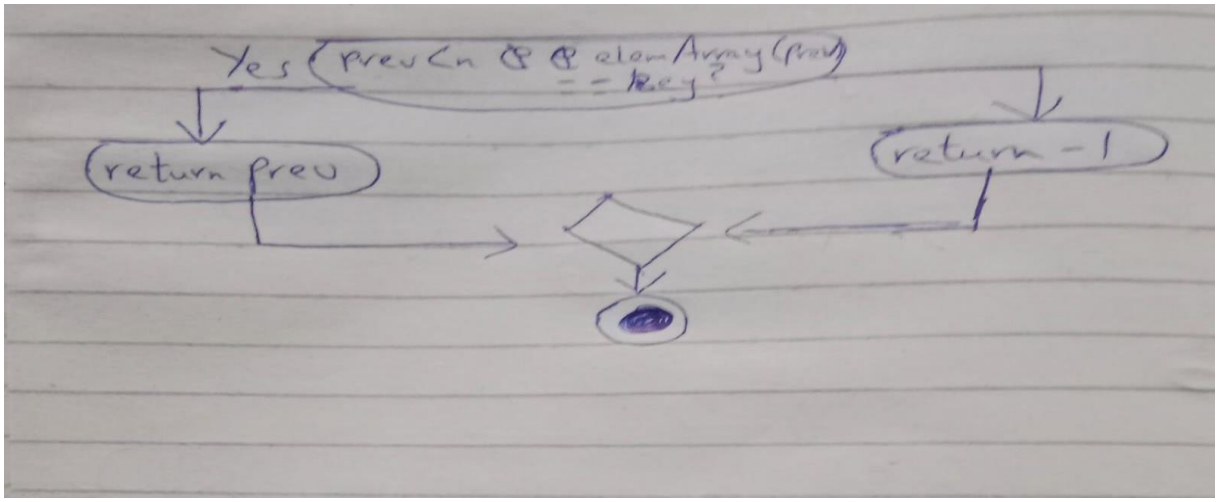
4. Control Flow Graphs (CFG)

4.1 Order class

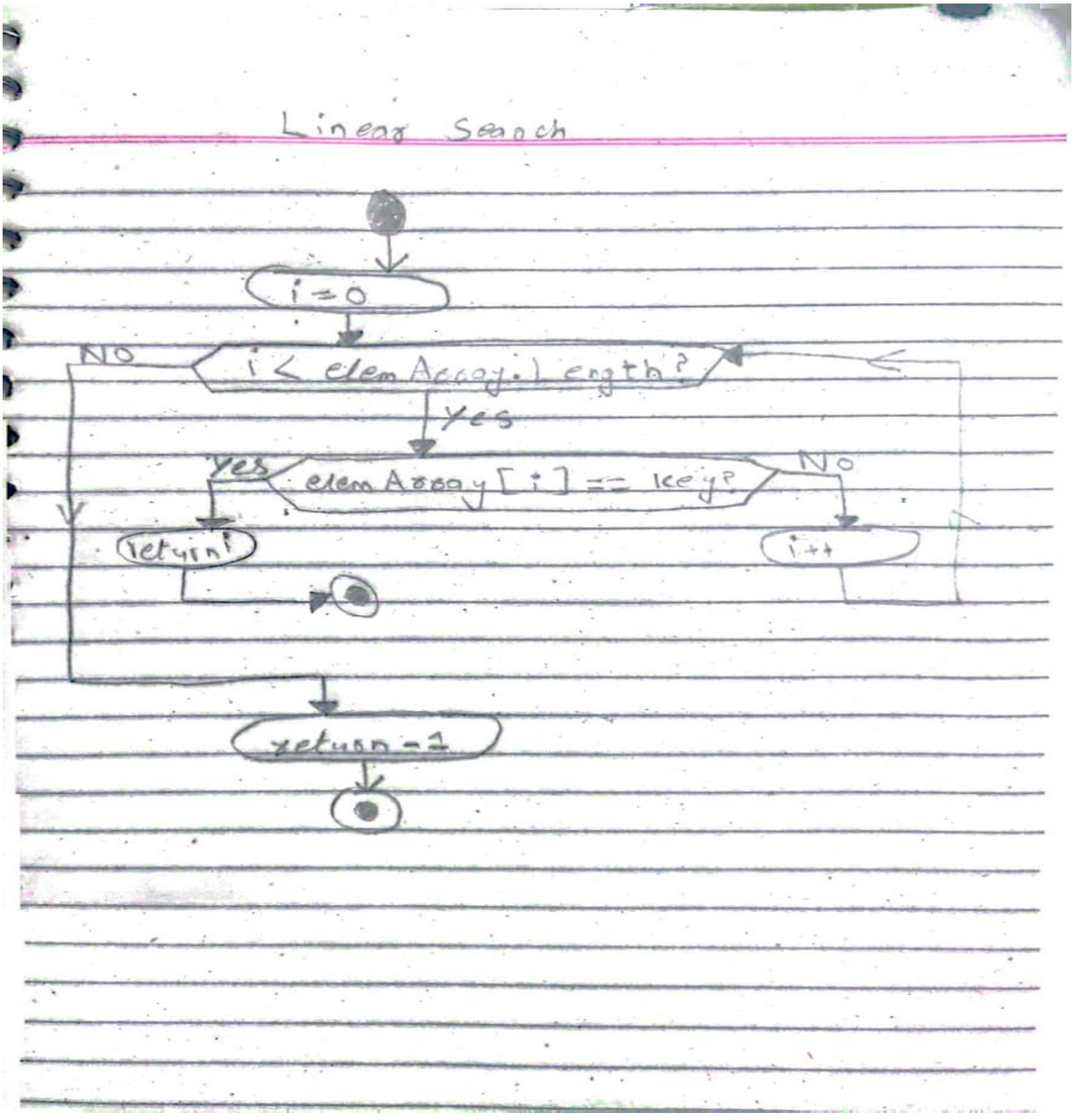


4.2 Jump Search

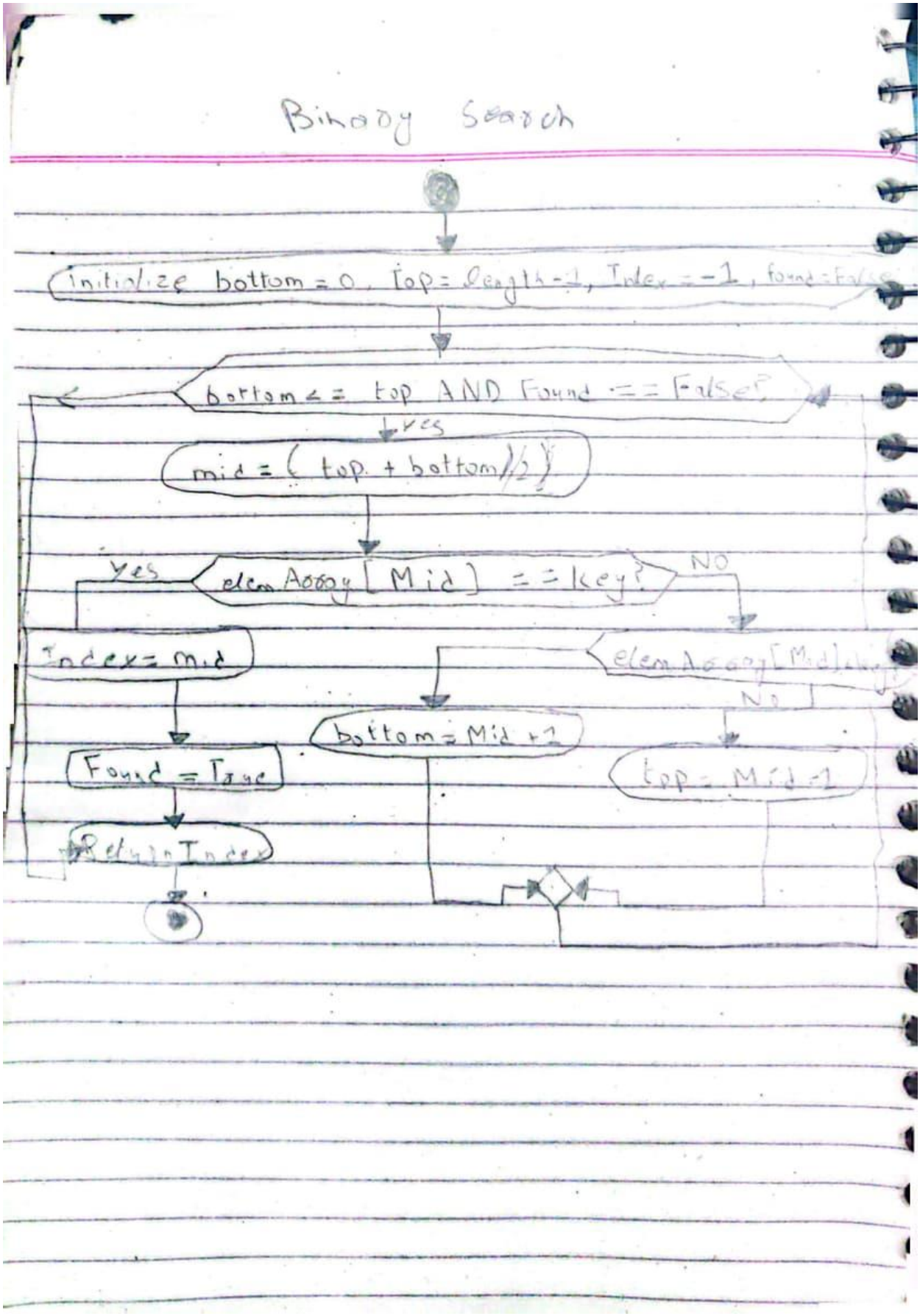




4.3 Linear Search

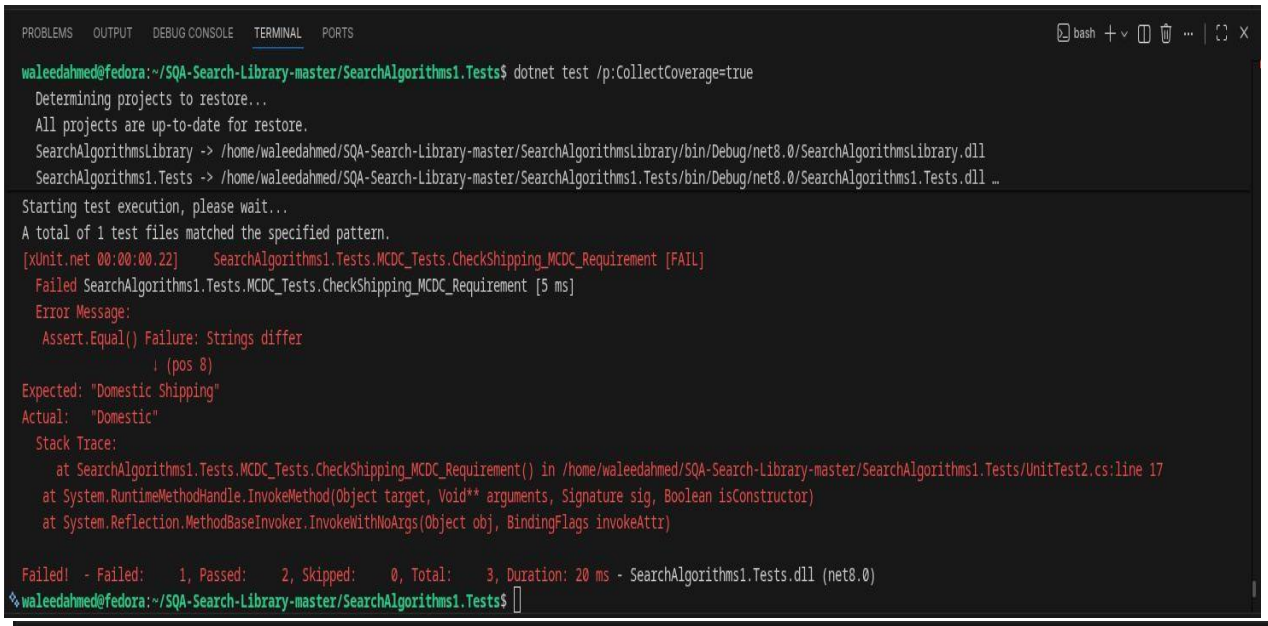


4.4 Binary Search



5. Technical Verification

5.1 Local xUnit Results

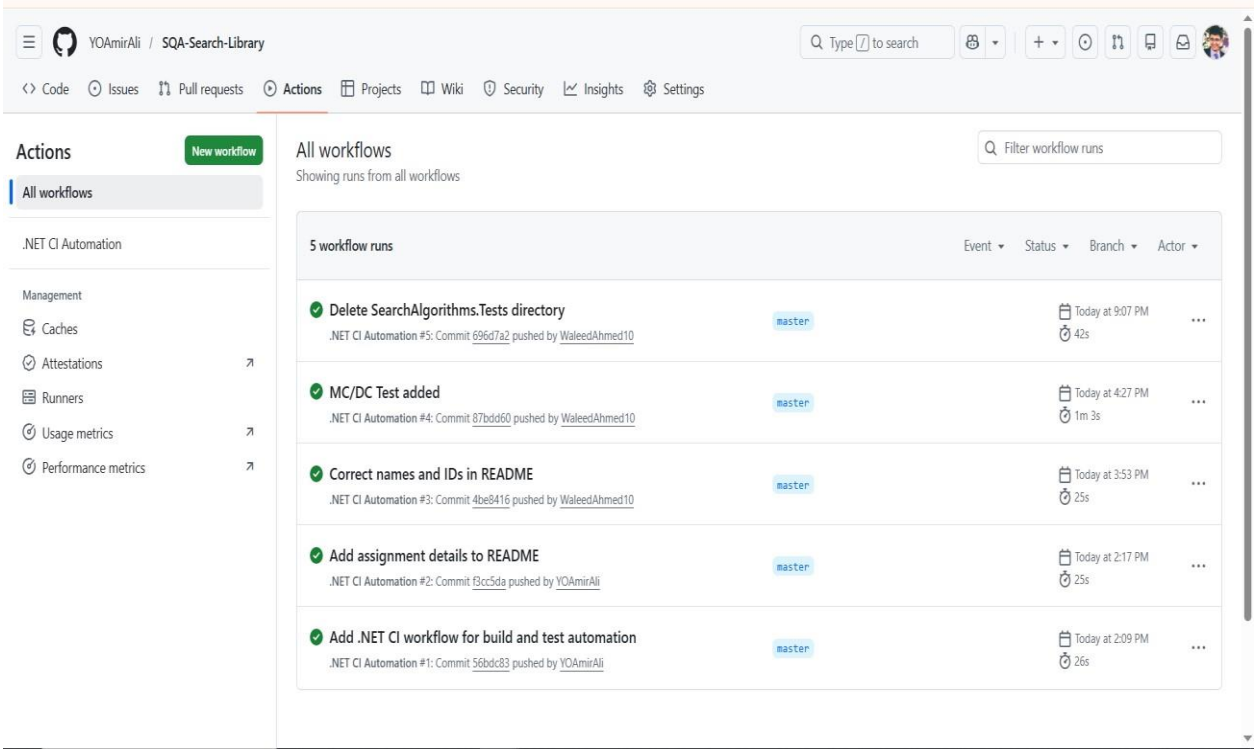




Initially, an integration error was caught where "Domestic" was returned instead of "Domestic Shipping." After alignment, a final test run confirmed that 4 out of 4 tests passed successfully.

5.2 Automated GitHub Actions Pipeline

The automation is fully operational on the master branch. Every push triggers the .NET CI Automation workflow, which handles dependencies, builds the solution, and executes the test suite in a clean cloud environment.



6. Conclusion

The project successfully demonstrates automated quality engineering. By combining manual path analysis (CFGs) with rigorous MC/DC test criteria and cloud-based automation, the SearchAlgorithmsLibrary is verified as stable, integrated, and ready for deployment.